Configure a Secure Server with SSL/TLS Using OpenSSL

Aqui comieinzo con creando una llave privada 2048-bit RSA. lo logre usando el siguiente syntax. **openssl genrsa -out /etc/ssl/private/myserver.key 2048 .**

```
debian@debian:~$ sudo openssl genrsa -out /etc/ssl/private/myserver.key 2048
[sudo] password for debian:
```

Here we generated a CSR.

```
debian@debian:~$ openssl req -new -key /etc/ssl/private/myserver.key -out /etc/ssl/private/myserver.c
sr
Could not open file or uri for loading private key from /etc/ssl/private/myserver.key
40D798CD2C7F0000:error:16000069:STORE routines:ossl_store_get0_loader_int:unregistered scheme:../cryp
to/store/store_register.c:237:scheme=file
40D798CD2C7F0000:error:8000000D:system library:file_open:Permission denied:../providers/implementatio
ns/storemgmt/file_store.c:267:calling stat(/etc/ssl/private/myserver.key)
debian@debian:~$ sudo openssl req -new -key /etc/ssl/private/myserver.key -out /etc/ssl/private/myser
ver.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:S
```

Aqui llenamos informacion general que va ir incluida con el certificado.

```
-----
Country Name (2 letter code) [AU]:MX
State or Province Name (full name) [Some-State]:Chihuahua
Locality Name (eg, city) []:Chihuahua
Organization Name (eg, company) [Internet Widgits Pty Ltd]:MiEmpresa
Organizational Unit Name (eg, section) []:IT
Common Name (e.g. server FQDN or YOUR name) []:mi-dominio.com
Email Address []:admin@mi-dominio.com
```

Aqui firmamos el CSR con mi llave privada para obtener un self-signed certificate por 365 days.

```
debian@debian:~$ sudo openssl x509 -req -in /etc/ssl/private/myserver.csr -signkey /etc/ssl/private/myserve
r.key -out /etc/ssl/private/myserver.crt -days 365
Certificate request self-signature ok
subject=C = MX, ST = Chihuahua, L = Chihuahua, O = MiEmpresa, OU = IT, CN = mi-dominio.com, emailAddress =
admin@mi-dominio.com
debian@debian:~$
```

Aqui aseguramos que el archivo apache ssl este configurado adecuadamente

```
debian@debian: ~
File  Edit  View  Search  Terminal  Help
  GNU nano 7.2      /etc/apache2/sites-available/default-ssl.conf
```sh
<IfModule mod_ssl.c>
    <VirtualHost _default_:443>
        ServerAdmin admin@mi-dominio.com
        ServerName mi-dominio.com

        DocumentRoot /var/www/html

        SSLEngine on
        SSLCertificateFile /etc/ssl/certs/myserver.crt
        SSLCertificateKeyFile /etc/ssl/private/myserver.key

        <FilesMatch "\.(cgi|shtml|phtml|php)$">
            SSLOptions +StdEnvVars
        </FilesMatch>
        <Directory /usr/lib/cgi-bin>
            SSLOptions +StdEnvVars
```

Aqui configuramos el Hosts file and mi maquina host.

```
hosts: Bloc de notas
Archivo  Edición  Formato  Ver  Ayuda
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#      102.54.94.97     rhino.acme.com          # source server
#       38.25.63.10     x.acme.com              # x client host

# localhost name resolution is handled within DNS itself.
#       127.0.0.1       mi-dominio.com
#       ::1             localhost
Línea 20, columna 33   100%   Windows (CRLF)   UTF-8
```
```